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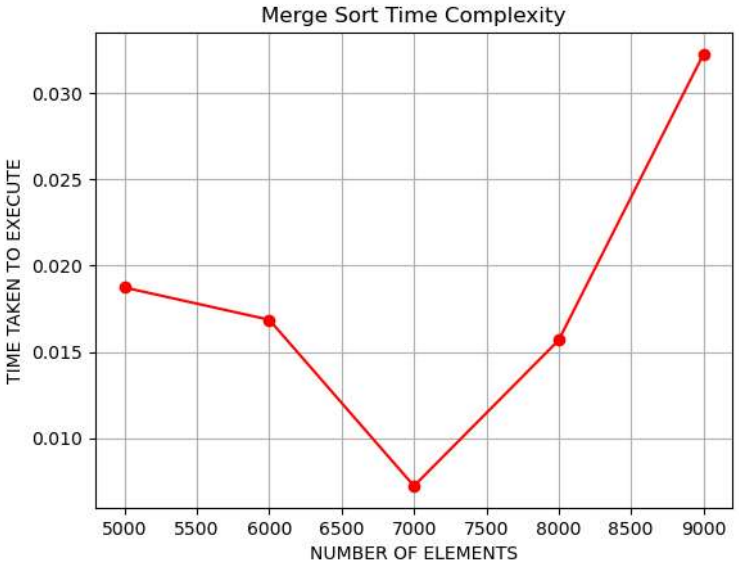
**EXPT.NO - 3**

**TITLE - MERGE SORT**

In [1]:

```python
import random
import time
import matplotlib.pyplot as plt
def merge_sort(arr):
    if len(arr) <= 1:
        return arr
    mid = len(arr)//2
    left = merge_sort(arr[:mid])
    right =  merge_sort(arr[mid:])
    return merge(left,right)
def merge(left,right):
    result =[]
    i = j = 0
    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1
    return result + left[i:] + right[j:]

#input for user
arr = list(map(int,input("ENTER THE ELEMENTS SEPERETED BY SPACE").split()))
print("INPUT array: ",arr)
start_time = time.time()
sorted_arr = merge_sort(arr)
end_time = time.time()
print("Sorted array:", sorted_arr)
print("Time taken to sort",end_time - start_time,"seconds")
print("==================================================================================")
#for n>5000
n_values = [5000, 6000,7000,8000,9000]
time_values=[]
for n in n_values:
    arr=[random.randint(1,9) for _ in range(n)]
    start_time=time.time()
    sorted_arr = merge_sort(arr)
    end_time = time.time()
    time_taken = end_time-start_time
    print("Time taken to sort",n,"elements",time_taken,"seconds")
    time_values.append(time_taken)
    print("==================================================================================")
#ploting the graph
plt.plot(n_values, time_values, color ='red',marker='o')
plt.xlabel("NUMBER OF ELEMENTS")
plt.ylabel("TIME TAKEN TO EXECUTE")
plt.title("Merge Sort Time Complexity")
plt.grid(True)
plt.show()
```

```
ENTER THE ELEMENTS SEPERETED BY SPACE3 435 32 6 86 8 712 874
INPUT array:  [3, 435, 32, 6, 86, 8, 712, 874]
Sorted array: [3, 6, 8, 32, 86, 435, 712, 874]
Time taken to sort 0.0 seconds
==================================================================================
Time taken to sort 5000 elements 0.01872706413269043 seconds
==================================================================================
Time taken to sort 6000 elements 0.01685786247253418 seconds
==================================================================================
Time taken to sort 7000 elements 0.007235288619995117 seconds
==================================================================================
Time taken to sort 8000 elements 0.015677213668823242 seconds
==================================================================================
Time taken to sort 9000 elements 0.03223228454589844 seconds
==================================================================================
```